PATIENT DATA ANALYSIS

Understanding patient wait times

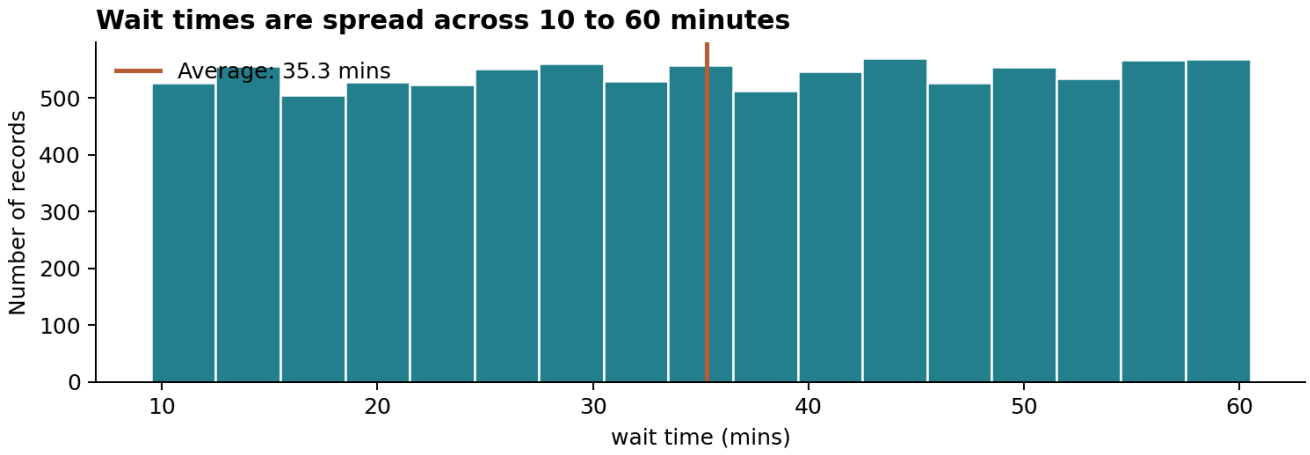
What I found and how well I could predict the wait

I analysed 9,216 patient records to understand whether gender, age, race, referral department and admission status were related to wait time. I then used regression to test whether the engineered features could predict how long a patient would wait.

9,216	**35.3 minutes**	**10 to 60 minutes**
patient records	average wait	observed range

My main finding

I found that average wait times were similar across the groups I compared. Differences between individual patients were much larger than the differences between group averages. My regression model also performed almost the same as simply predicting the average wait time for everyone.



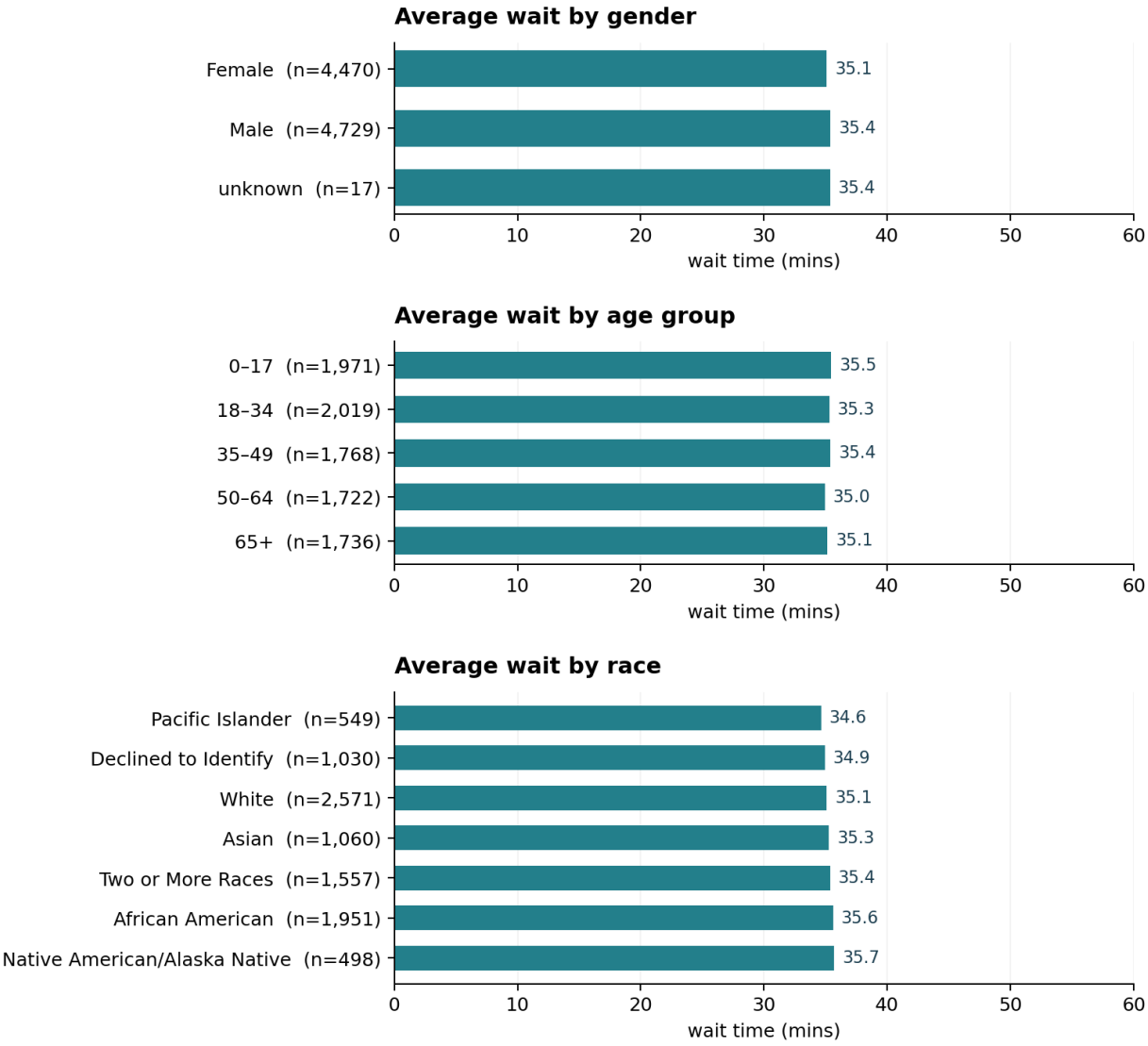
I found a median wait of 35 minutes. Half of the records had waits between 23 and 48 minutes. The chart shows that waits were spread across much of the observed range, rather than tightly grouped around the average.

How I prepared the data

I converted admission dates and times, removed hyphens from patient IDs and renamed the merged name column to pt name. I removed Patient Satisfaction Score because it was not needed for this analysis and 72.69% of its values were missing. I labelled 17 unclear gender entries as unknown because I did not want to assume a gender. I treated None under referral as no referral.

The group averages were very similar

I compared average wait times and reviewed the spread within each group. In these charts, n is the number of records in the group. I kept the same 0 to 60 minute scale to show the size of the differences fairly.



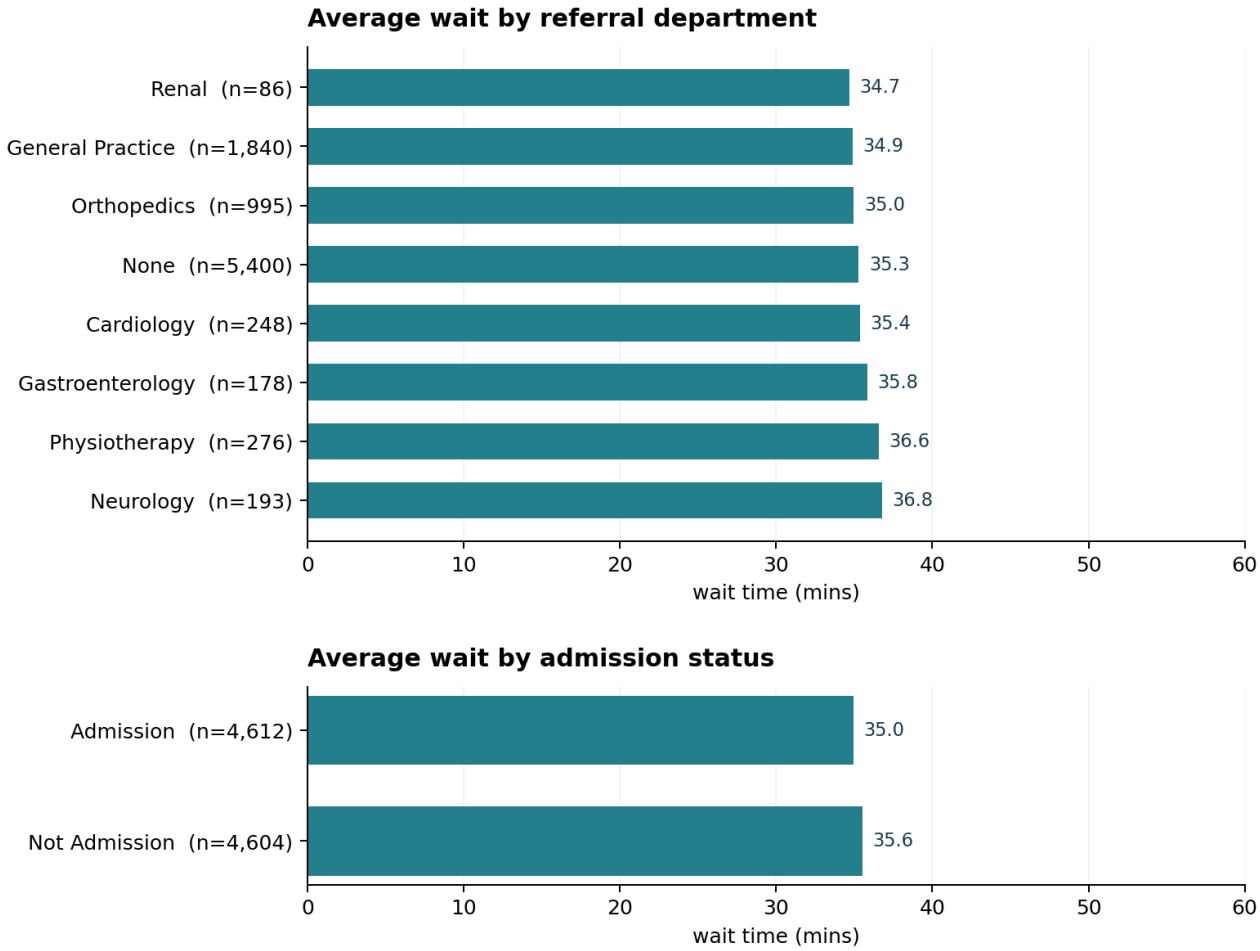
What I learned

I found average waits of 35.1 minutes for Female, 35.4 for Male and 35.4 for unknown. The unknown group had only 17 records, so I treated its average with more caution. Age-group averages ranged from 35.0 to 35.5 minutes. The relationship between age and wait time was almost zero.

Race-group averages ranged from 34.6 to 35.7 minutes. I did not interpret these small differences as evidence that race or gender causes a change in waiting time. The distributions overlapped substantially, and these comparisons did not account for other possible influences.

Department differences were small

I found the largest difference in group averages when comparing referral departments. However, even this relationship explained only about 0.066% of the variation in wait time, which is a very small share.



What I learned

Average waits ranged from 34.7 minutes for Renal to 36.8 minutes for Neurology, a difference of about 2.1 minutes. I also noted that some departments had relatively few records, including Renal with 86. I would not treat this ranking as evidence that one department provides faster service.

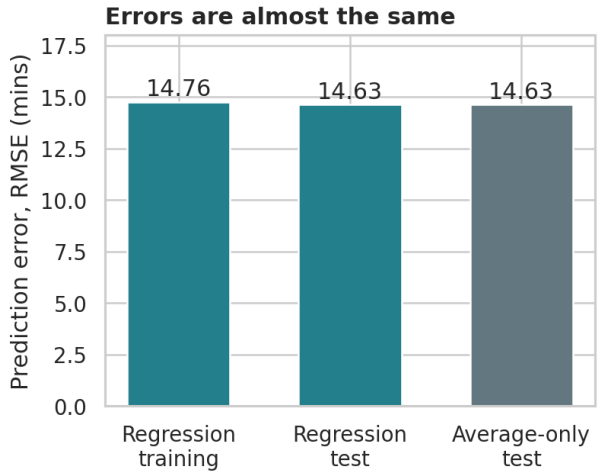
Records labelled Admission had an average wait of 35.0 minutes, compared with 35.6 minutes for Not Admission. This difference was small. I used admission status for this comparison but excluded it from prediction because I could not confirm whether it was known before the patient finished waiting.

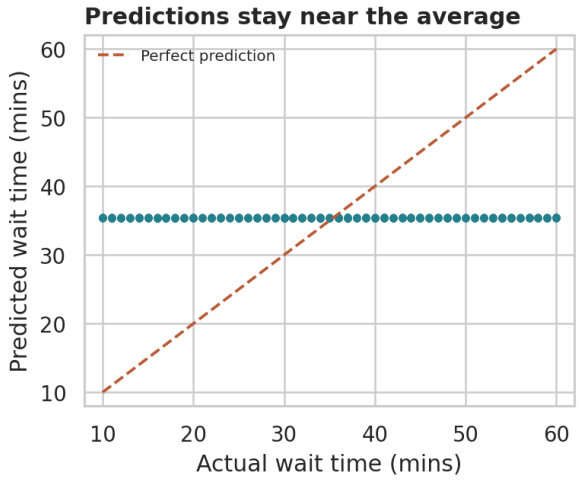
Across all five comparisons, each factor accounted for less than 0.1% of the differences in wait time on its own. I concluded that these characteristics alone did not explain much of the waiting-time variation in this dataset.

The model had limited accuracy

I created age, calendar, time and referral features, and converted categories into 0/1 columns. I excluded patient names and IDs from the predictors.

I used 7,372 records for training and held back 1,844 for testing. I compared linear regression with Ridge regression, a version that limits large model coefficients. I selected Ridge using five rounds of validation within the training data, then evaluated it on the held-out test set.





How I read these results

I measured prediction error using RMSE, which gives more weight to larger mistakes. Lower is better. My model had a training error of 14.76 minutes and a test error of 14.63 minutes. The average-only prediction also had a test error of 14.63 minutes. The difference was less than 0.01 minute, so the improvement was negligible.

I checked training and test errors and learning curves for overfitting. I found no large performance gap, but the model still had little predictive value. Its test R-squared was about -0.003, indicating almost no ability to explain individual differences. Predictions stayed near 35 minutes even when actual waits were much shorter or longer.

My conclusion

I could describe the average wait in this dataset, but I could not reliably predict individual waiting times from these features. I would need additional information, such as urgency, staffing levels or patient volume, to investigate whether those factors explain more of the variation. Those factors were not available in this analysis.

I based these findings on one random train/test split. I have not established performance for future admissions, and I have not shown that any characteristic causes a change in wait time.

Sources: Kaggle_HealthData.csv and my cleaning, exploratory analysis, feature engineering and regression notebooks. Wait time is measured in minutes. Group charts show averages, not adjusted or causal effects.